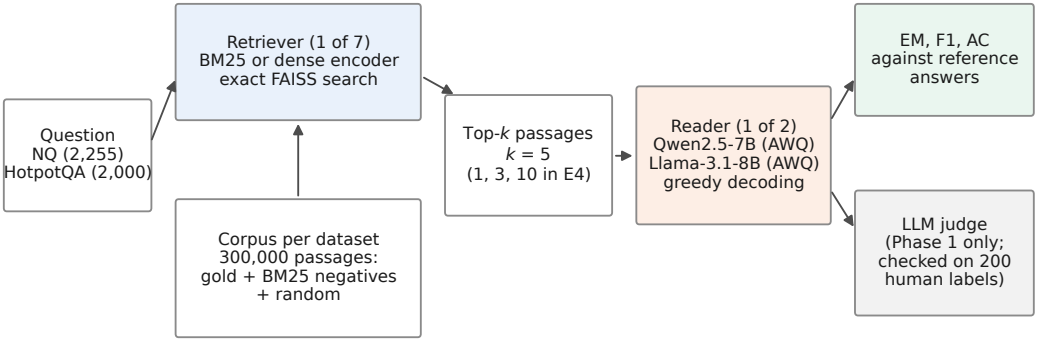cost measured at this stage: encoding throughput, query latency, index size, GPU memory